VISUALISATIONS UNIVARIÉES ET BIVARIÉES

OBJECTIFS D'APPRENTISSAGE

- Sélectionner la visualisation adaptée selon le type de variable et l'objectif analytique
- Maîtriser les principes de data visualization (Tufte, Cleveland)
- Créer des graphiques avec matplotlib et seaborn
- Identifier les pièges classiques de la visualisation

TAXONOMIE DES VISUALISATIONS

La taxonomie des visualisations classe les graphiques selon **le type de données** et **l'objectif analytique**.

- **Univariée** : Une seule variable (ex. histogramme, diagramme en barres).
- **Bivariée** : Deux variables (ex. nuage de points, boxplot par groupe).
- **Multivariée** : Plus de deux variables (ex. scatterplot matrix, coordonnées parallèles).
- **Hiérarchique, spatiale, temporelle** : selon le type de données

QUARTET D'ANSCOMBE

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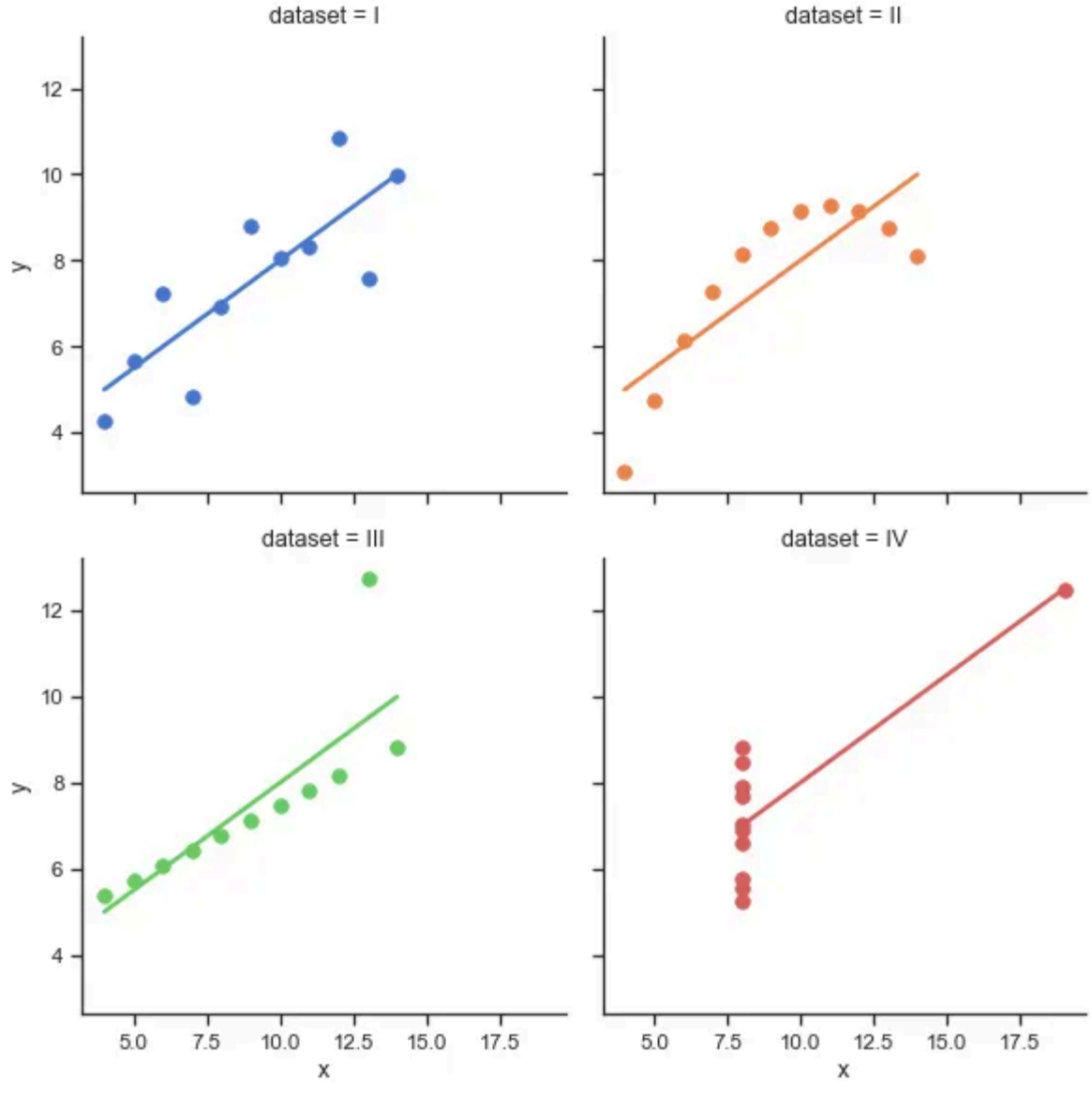
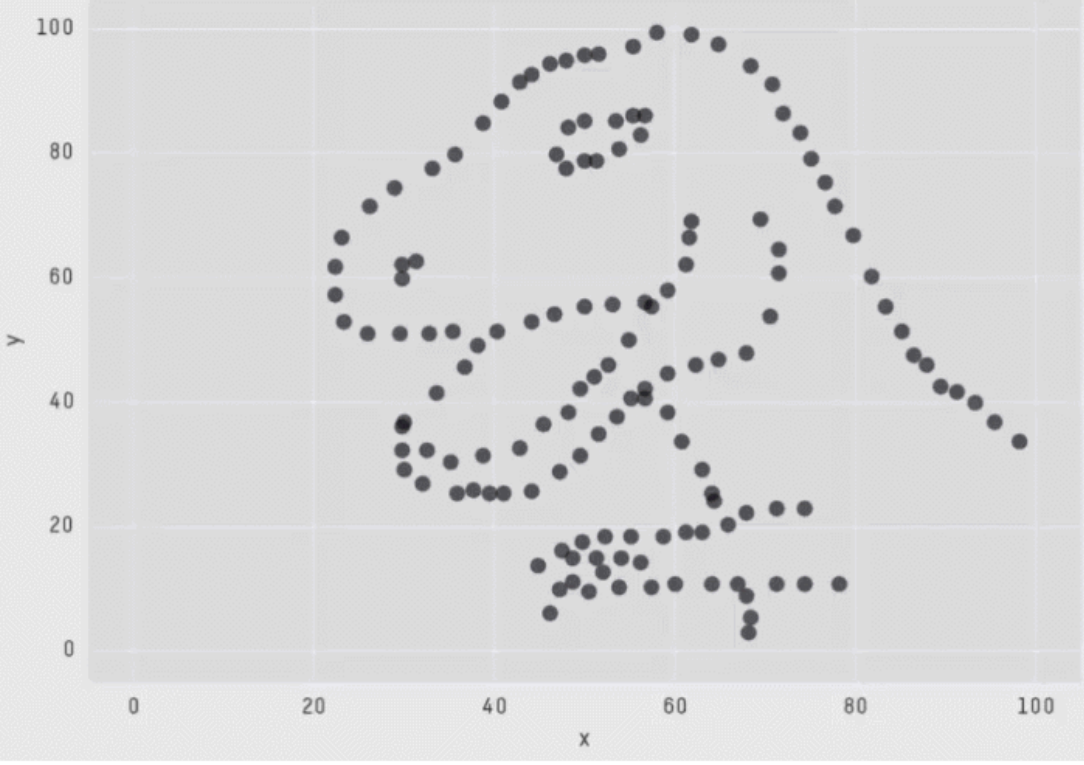
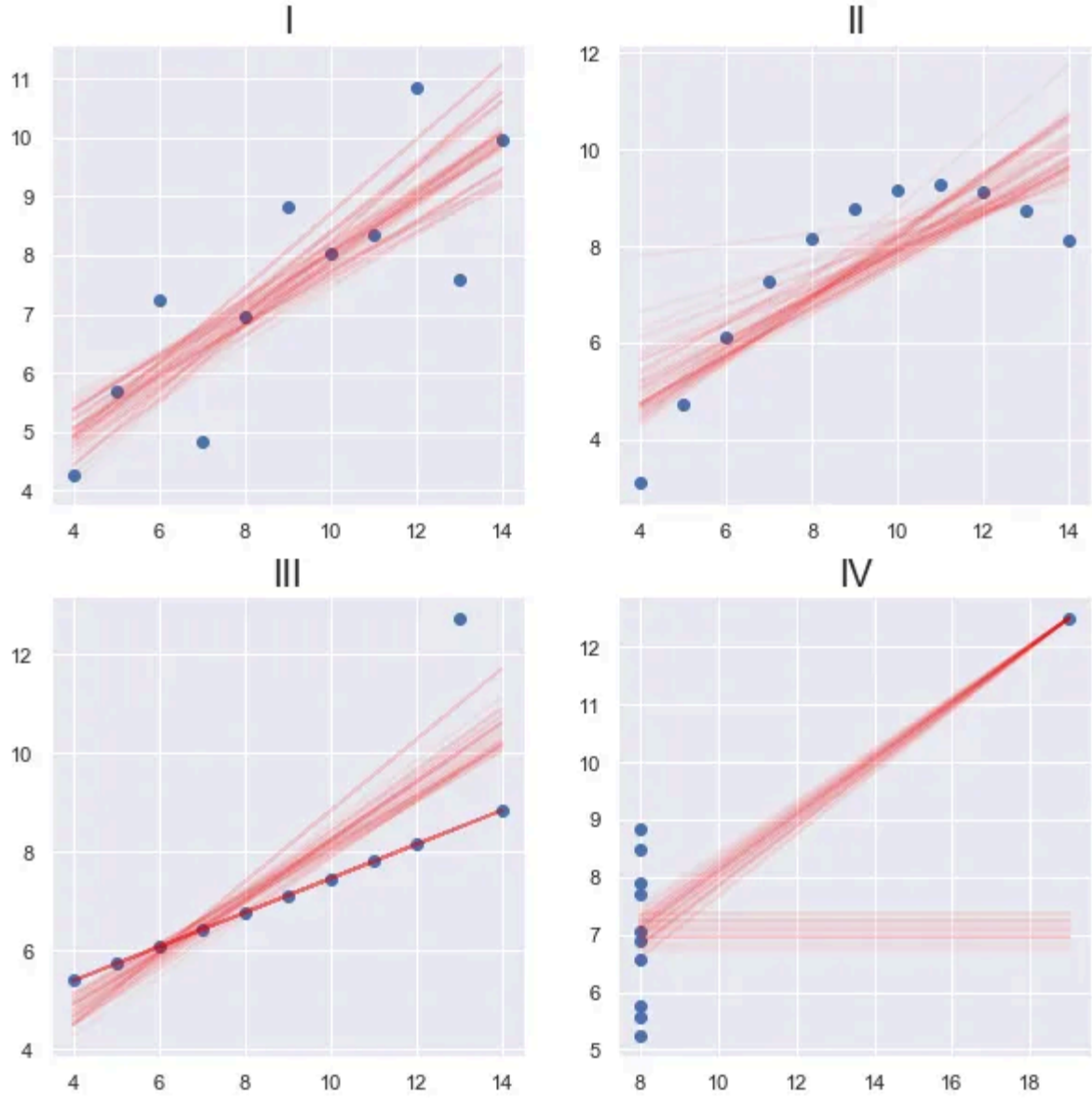


LE DINOSAURE D'ANSCOMBE



X Mean: 54.2659224
Y Mean: 47.8313999
X SD : 16.7649829
Y SD : 26.9342120
Corr. : -0.0642526


I
II
III
IV

SÉLECTIONNER LA VISUALISATION ADAPTÉE

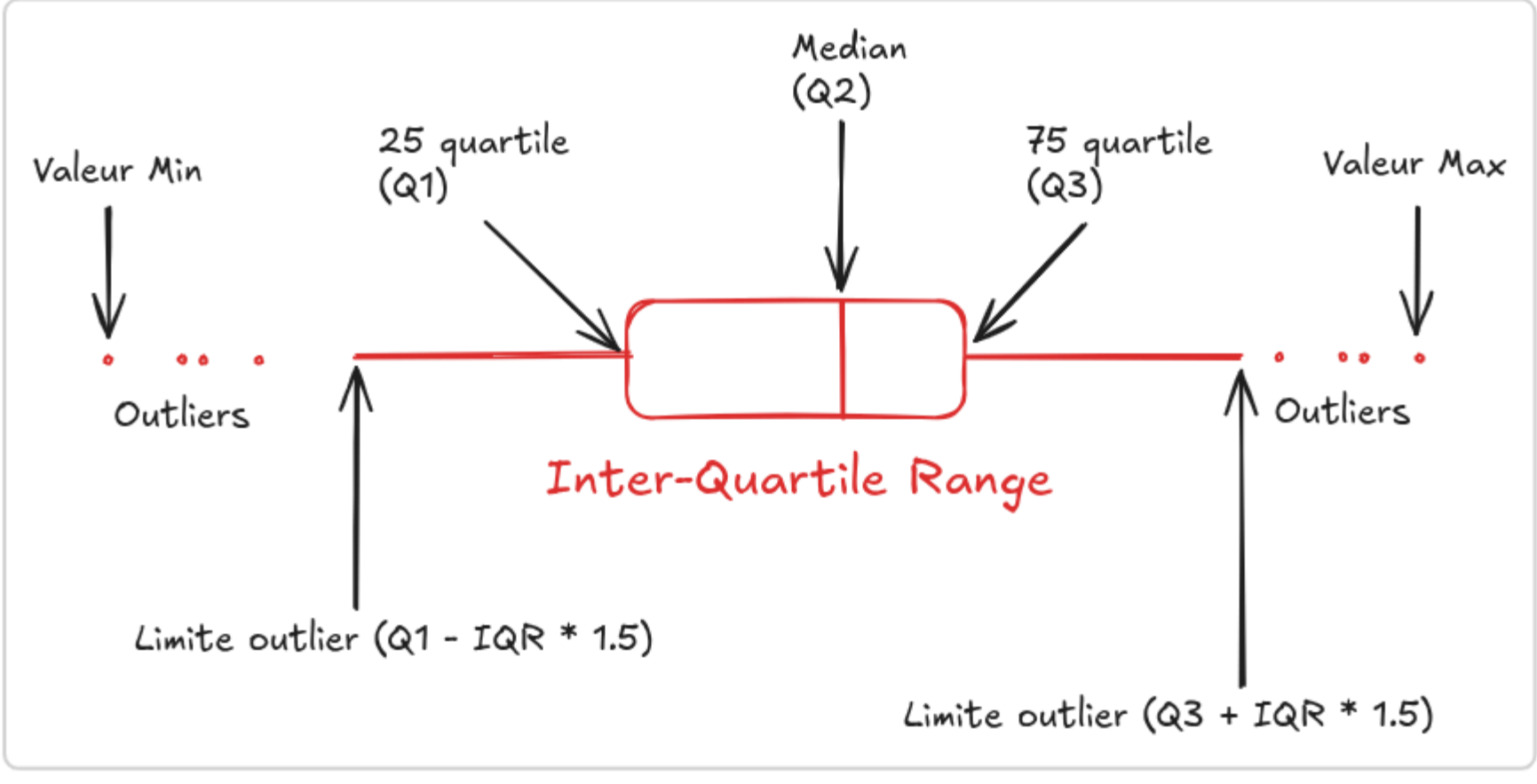
Le choix du graphique dépend du **type de variable** (nominale, ordinale, quantitative) et de **l'objectif analytique**.

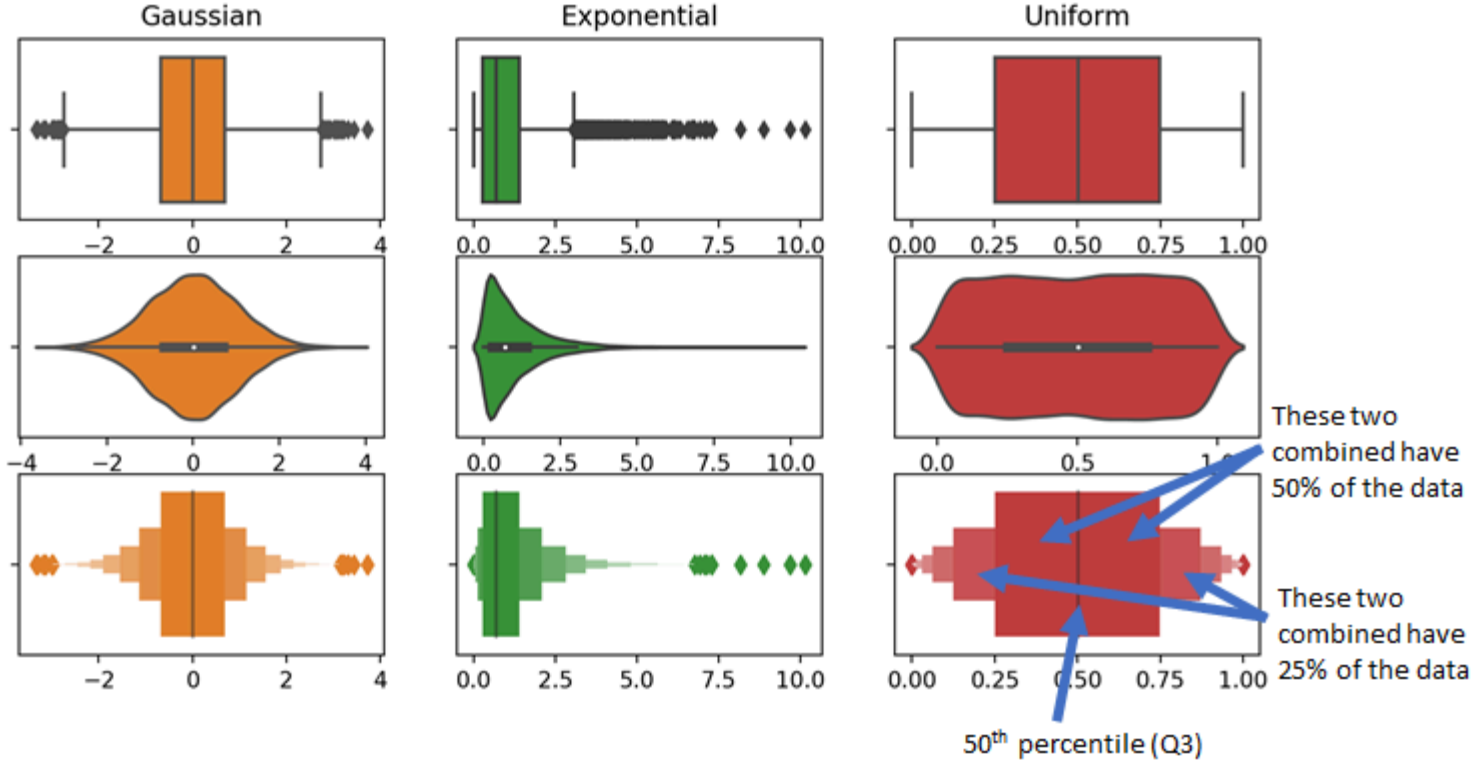
- **Variables qualitatives** : barres (comparaison), camembert (parts, si peu de catégories)
- **Variables quantitatives** : histogramme (distribution), boxplot (résumé), KDE (lissage), scatterplot (relation)
- **Mixtes** : boxplot par groupe, barres avec erreurs

COMPRENDRE LE BOXPLOT & VARIATIONS

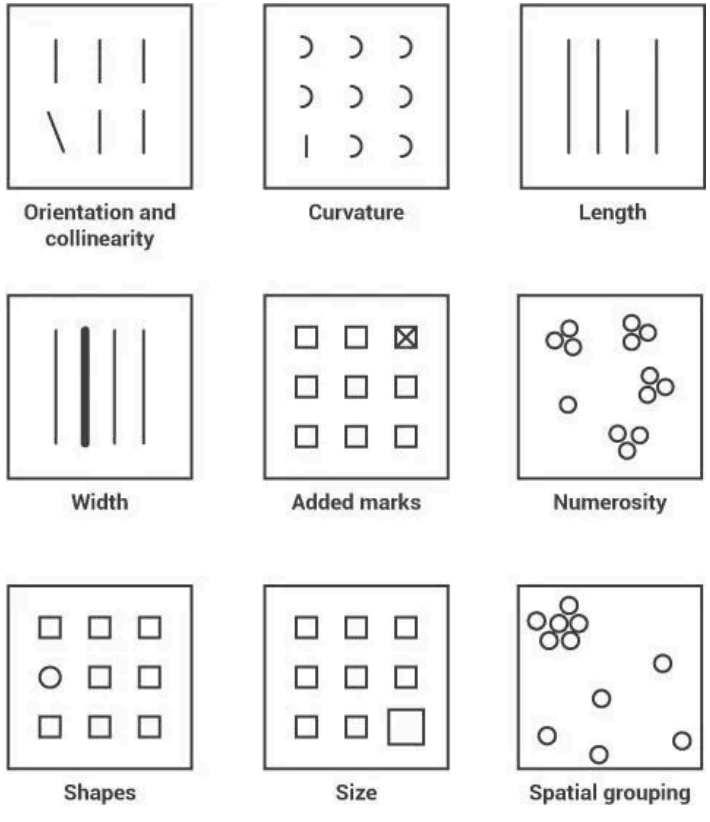
Pour visualiser les statistiques d'une variable on peut utiliser 3 graphiques différents: le **boxplot**, le **boxenplot**, le **violinplot**

Boxplot




Gaussian
Exponential
Uniform
These two combined have 50% of the data
These two combined have 25% of the data
50th percentile (Q3)

PRÉ-ATTENTIVE PROCESSING



- Attributs détectés en <250ms, inconsciemment
- Exemples : couleur, taille, orientation, forme, mouvement
- Guide efficacement l'attention dans un graphique

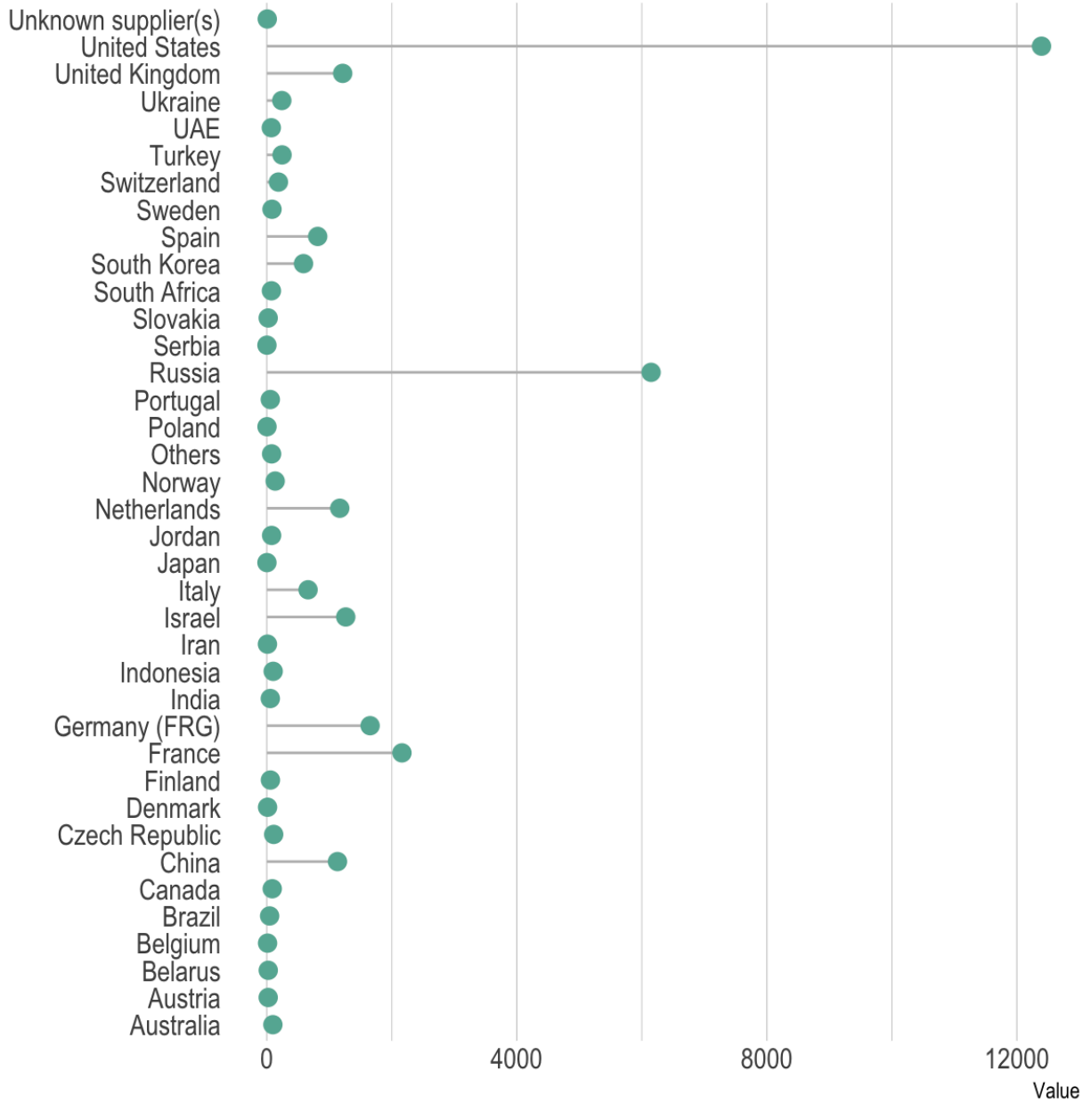
PRINCIPES DE LA GESTALT

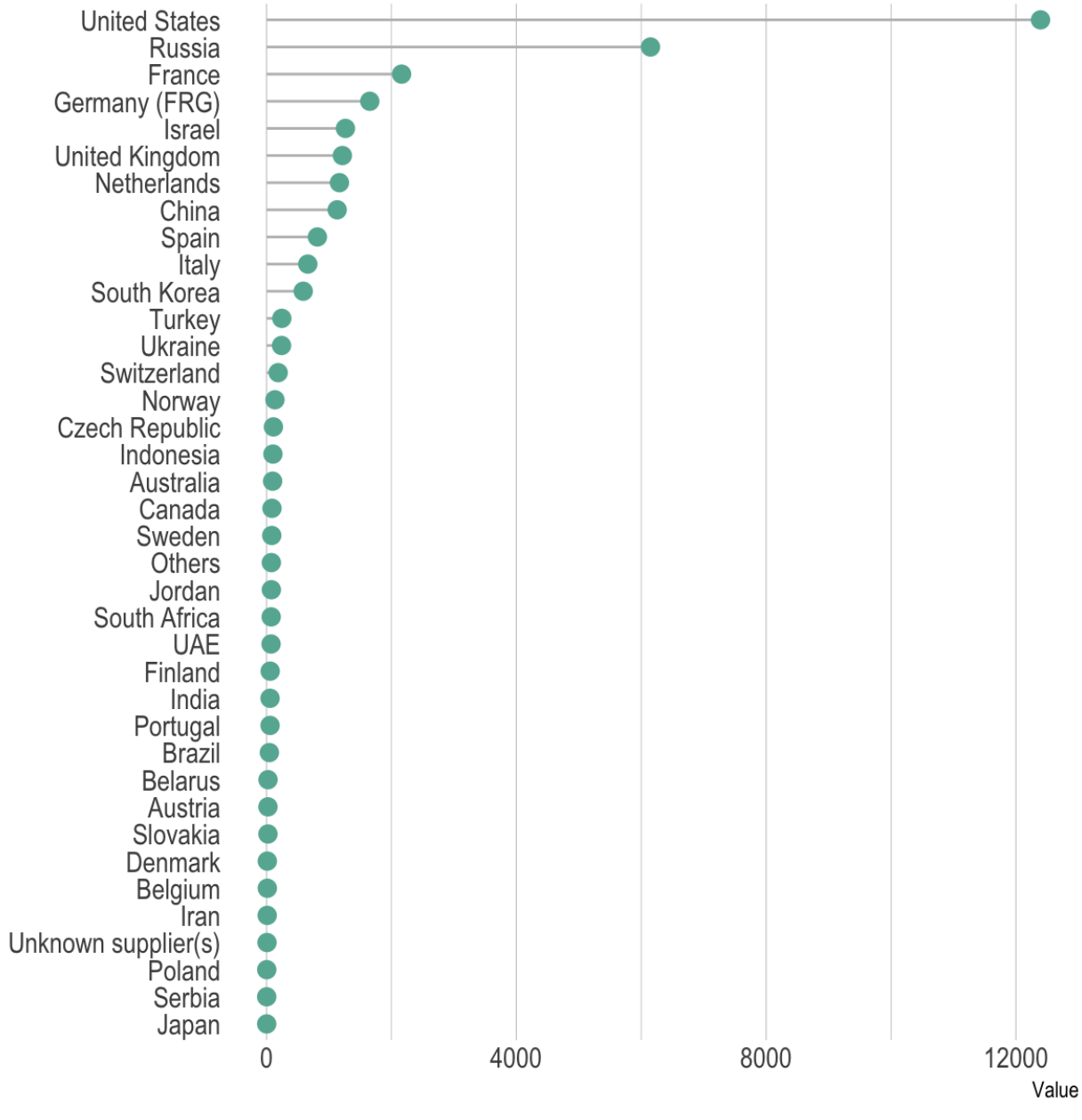
ANNOTATIONS CONTEXTUELLES

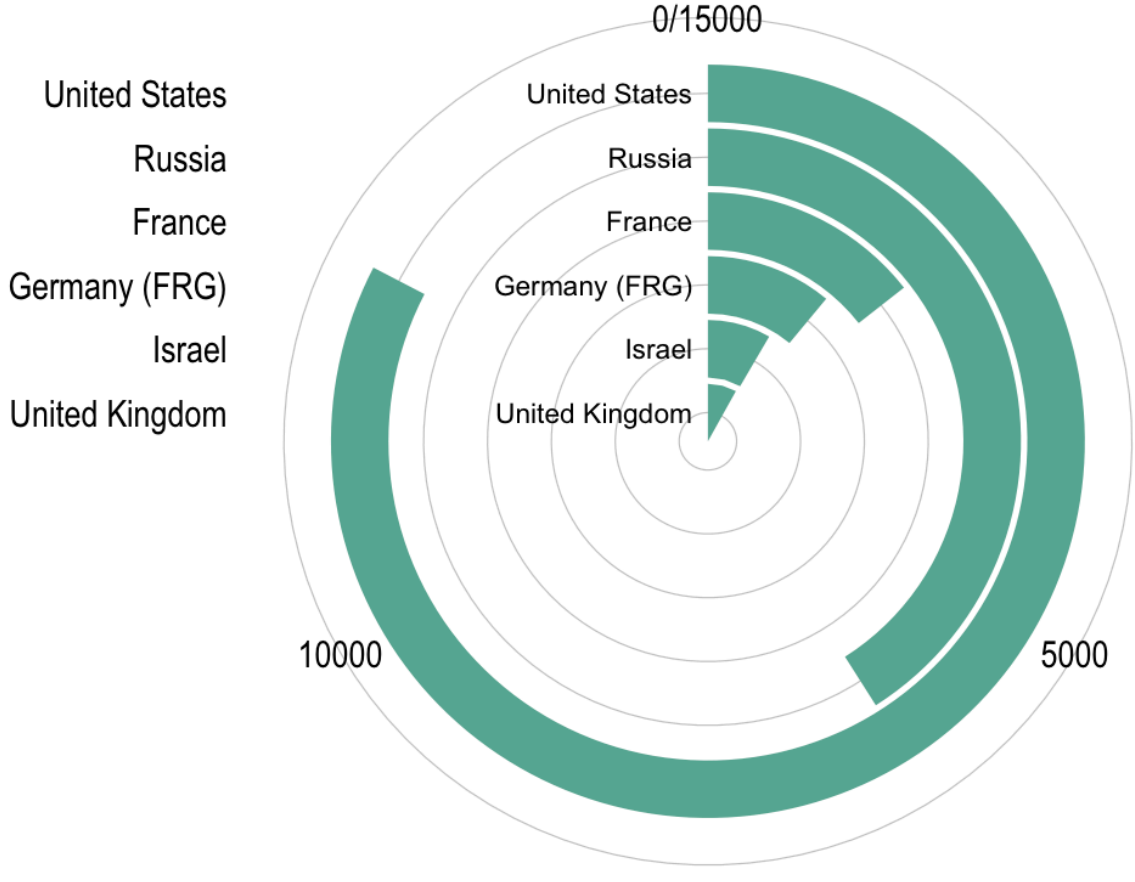
- **Lignes de seuil** : ex. ligne horizontale à 0 pour montrer la rentabilité.
- **Moyennes ou médianes** : lignes pointillées pour comparer les observations à la tendance centrale.
- **Événements marquants** : flèches ou textes pour marquer un lancement produit, une crise, etc.
- `plt.axhline()` pour une moyenne, `plt.text()` ou `plt.annotate()` pour du texte.

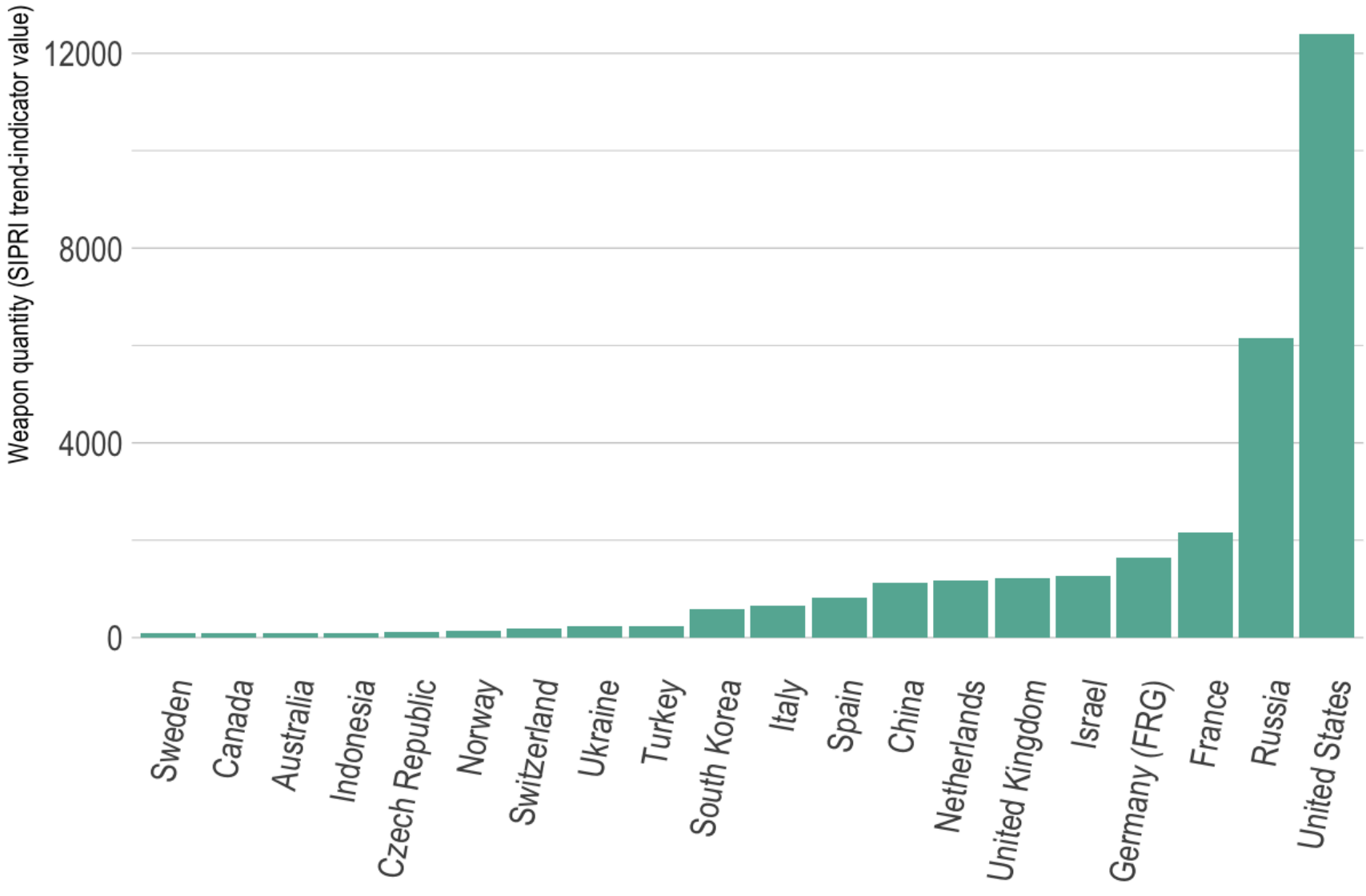
QUELQUES ERREURS COURANTES

EN 2017 ?

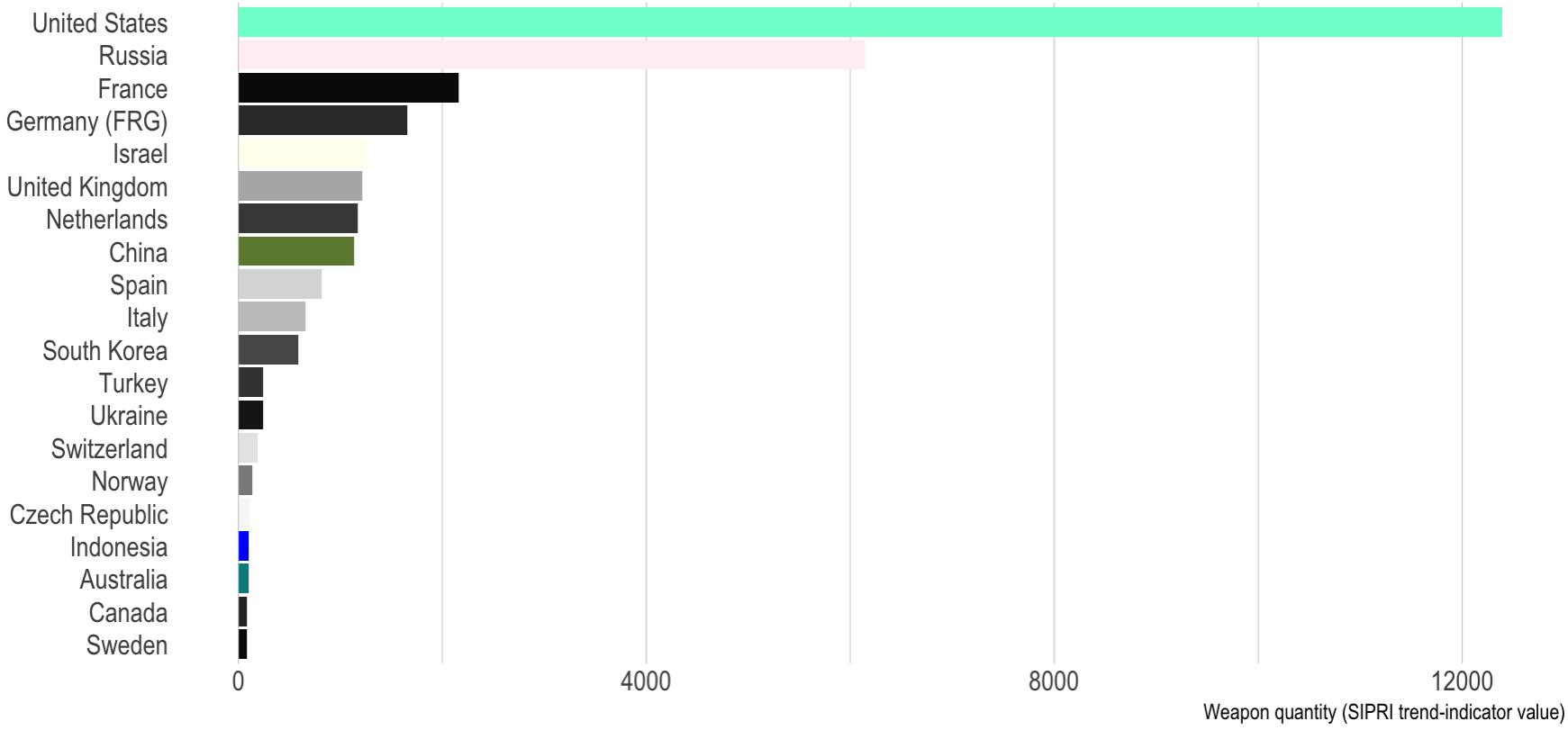

Unknown supplier(s)
United States
United Kingdom
Ukraine
UAE
Turkey
Switzerland
Sweden
Spain
South Korea
South Africa
Slovakia
Serbia
Russia
Portugal
Poland
Others
Norway
Netherlands
Jordan
Japan
Italy
Israel
Iran
Indonesia
India
Germany (FRG)
France
Finland
Denmark
Czech Republic
China
Canada
Brazil
Belgium
Belarus
Austria
Australia
0
4000
8000
12000
Value

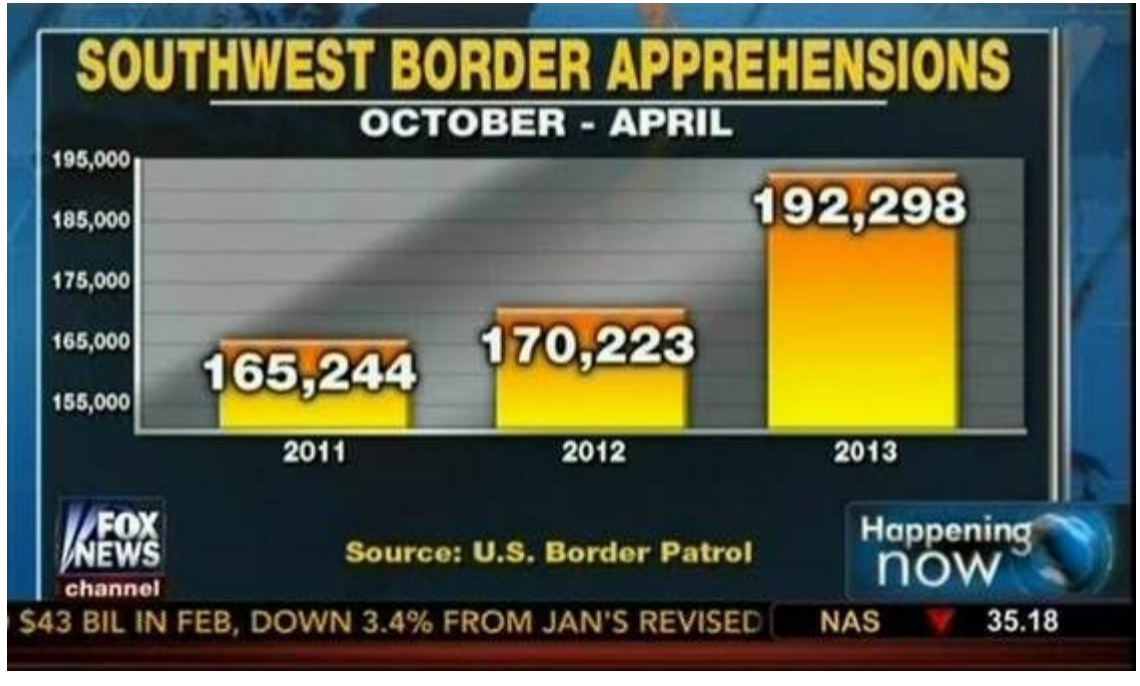

United States
Russia
France
Germany (FRG)
Israel
United Kingdom
Netherlands
China
Spain
Italy
South Korea
Turkey
Ukraine
Switzerland
Norway
Czech Republic
Indonesia
Australia
Canada
Sweden
Others
Jordan
South Africa
UAE
Finland
India
Portugal
Brazil
Belarus
Austria
Slovakia
Denmark
Belgium
Iran
Unknown supplier(s)
Poland
Serbia
Japan
0
4000
8000
12000
Value


0/15000
United States
Russia
France
Germany (FRG)
Israel
United Kingdom
United States
Russia
France
Germany (FRG)
Israel
United Kingdom
10000
5000

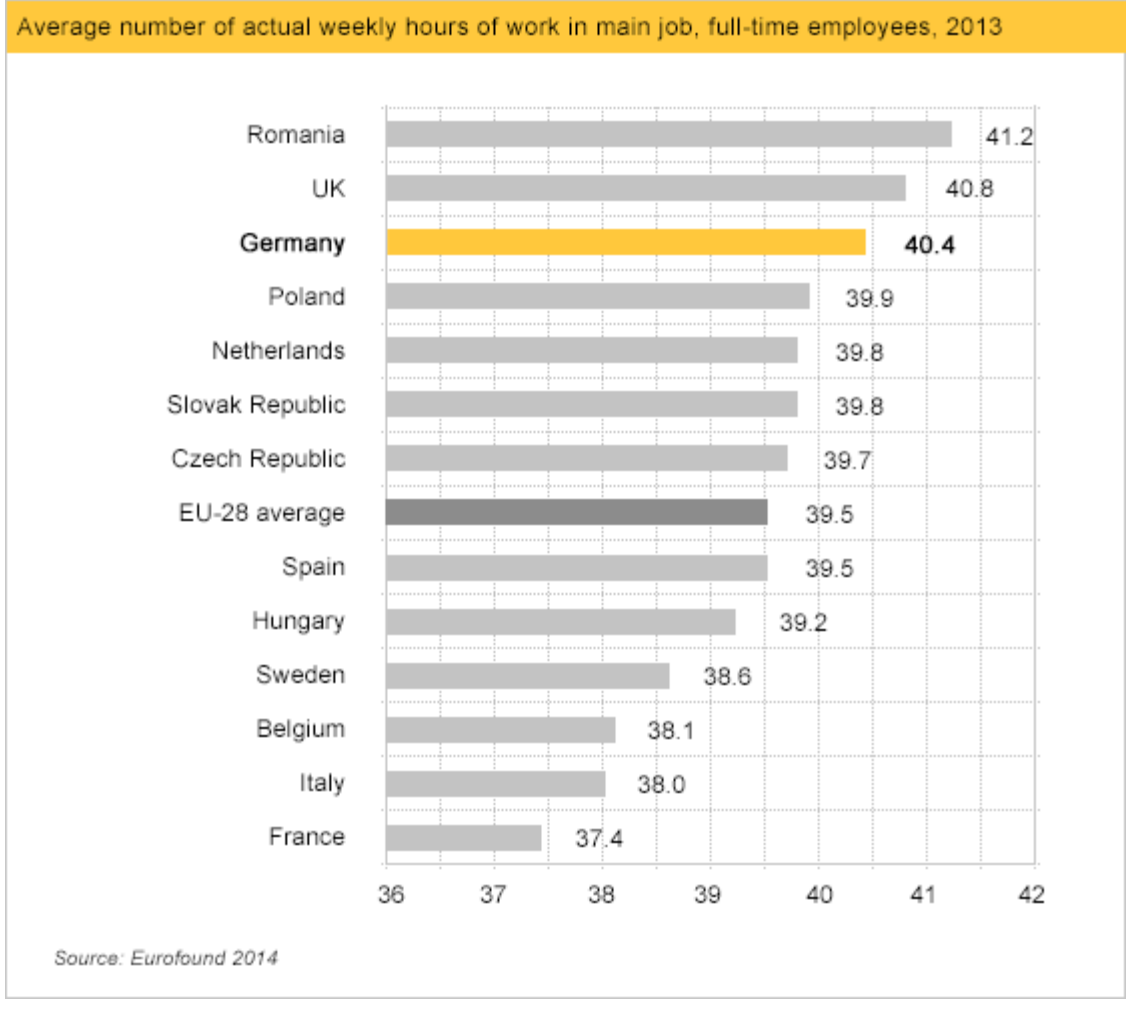

Weapon quantity (SIPRI trend-indicator value)
12000
8000
4000
0
Sweden
Canada
Australia
Indonesia
Czech Republic
Norway
Switzerland
Ukraine
Turkey
South Korea
Italy
Spain
China
Netherlands
United Kingdom
Israel
Germany (FRG)
France
Russia
United States

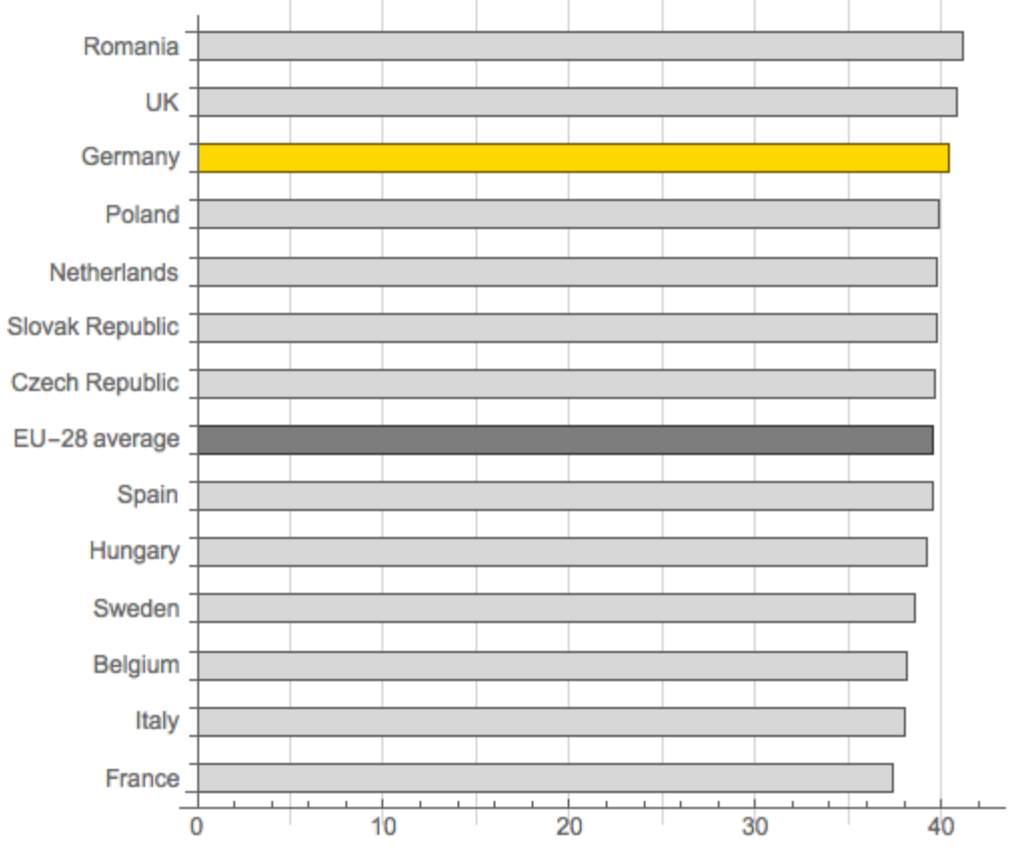
QUE REPRÉSENTENT LES COULEURS ?




SOUTHWEST BORDER APPREHENSIONS
OCTOBER - APRIL
195,000
185,000
175,000
165,000
155,000
165,244
170,223
192,298
2011
2012
2013
FOX NEWS channel
Source: U.S. Border Patrol
Happening now
$43 BIL IN FEB, DOWN 3.4% FROM JAN'S REVISED
NAS
35.18

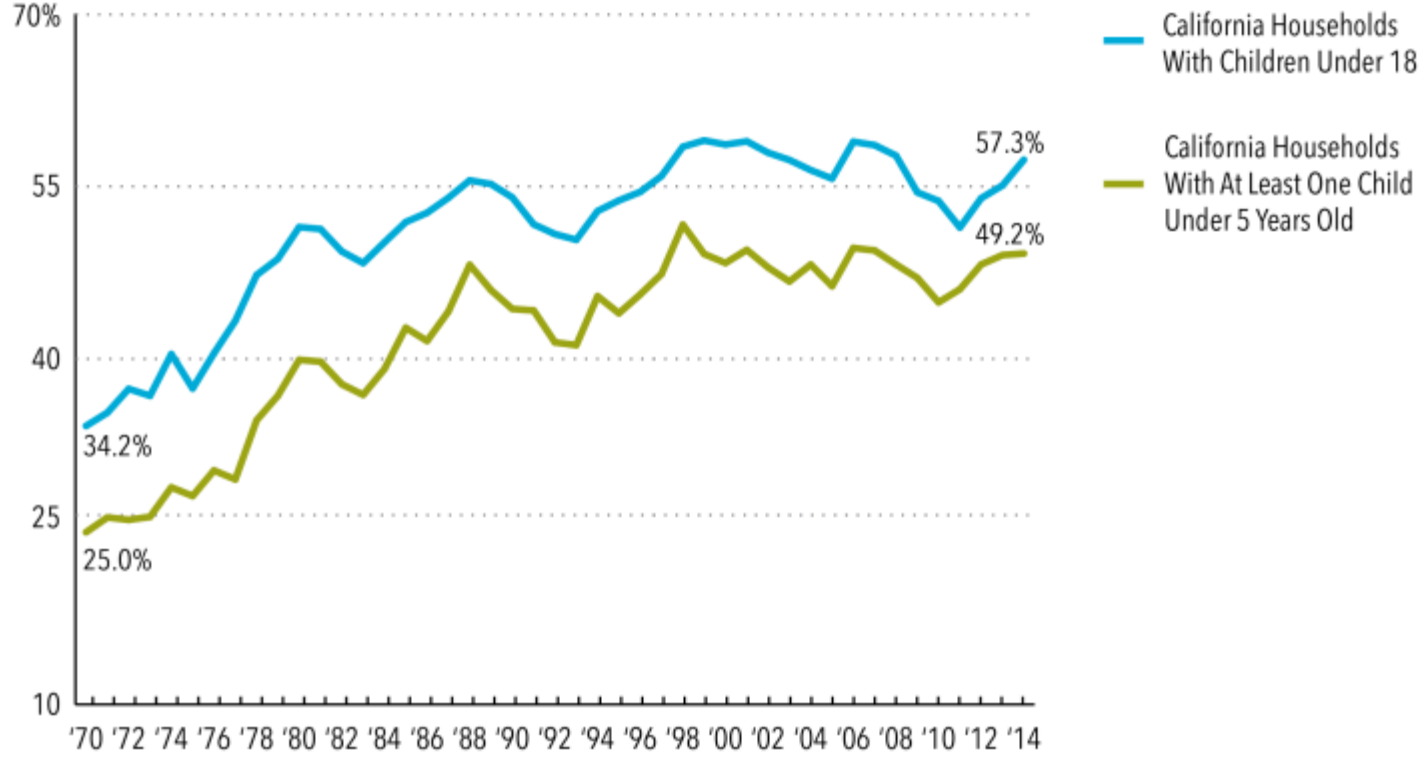
DE ZÉRO EN HÉRO




Romania
UK
Germany
Poland
Netherlands
Slovak Republic
Czech Republic
EU–28 average
Spain
Hungary
Sweden
Belgium
Italy
France
0
10
20
30
40

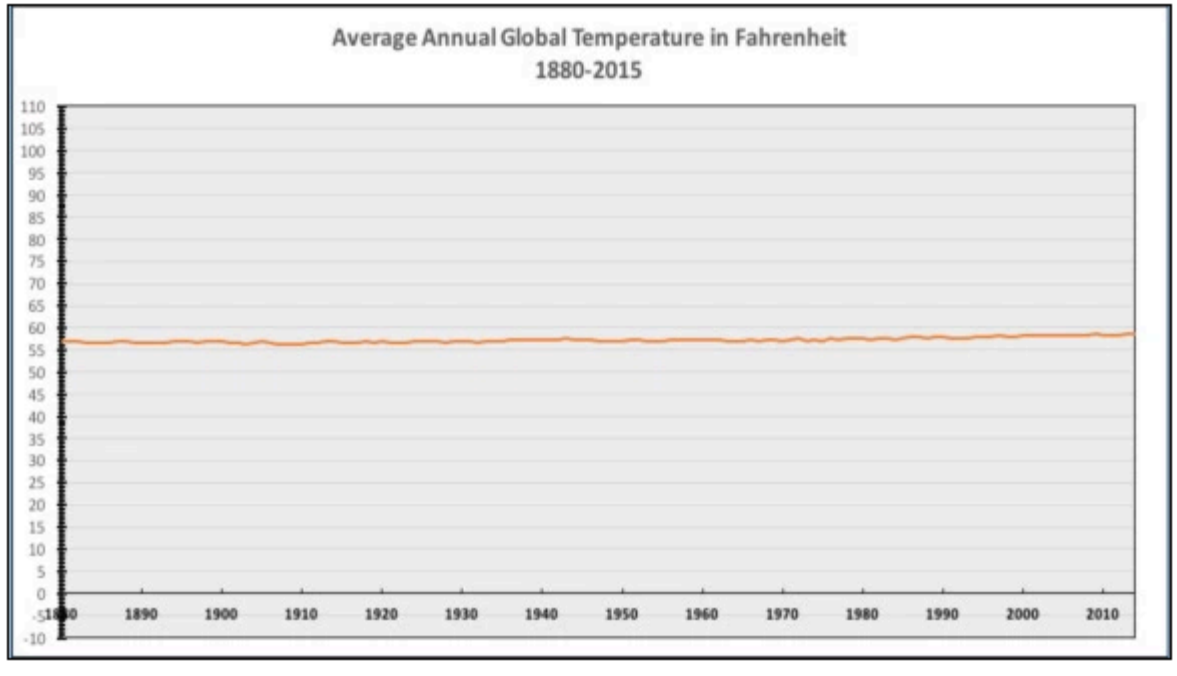
More California Households Have All Parents Working, Making Access to Child Care an Important Priority

Percentage of California Households Where All Parents Work, 1970 to 2014



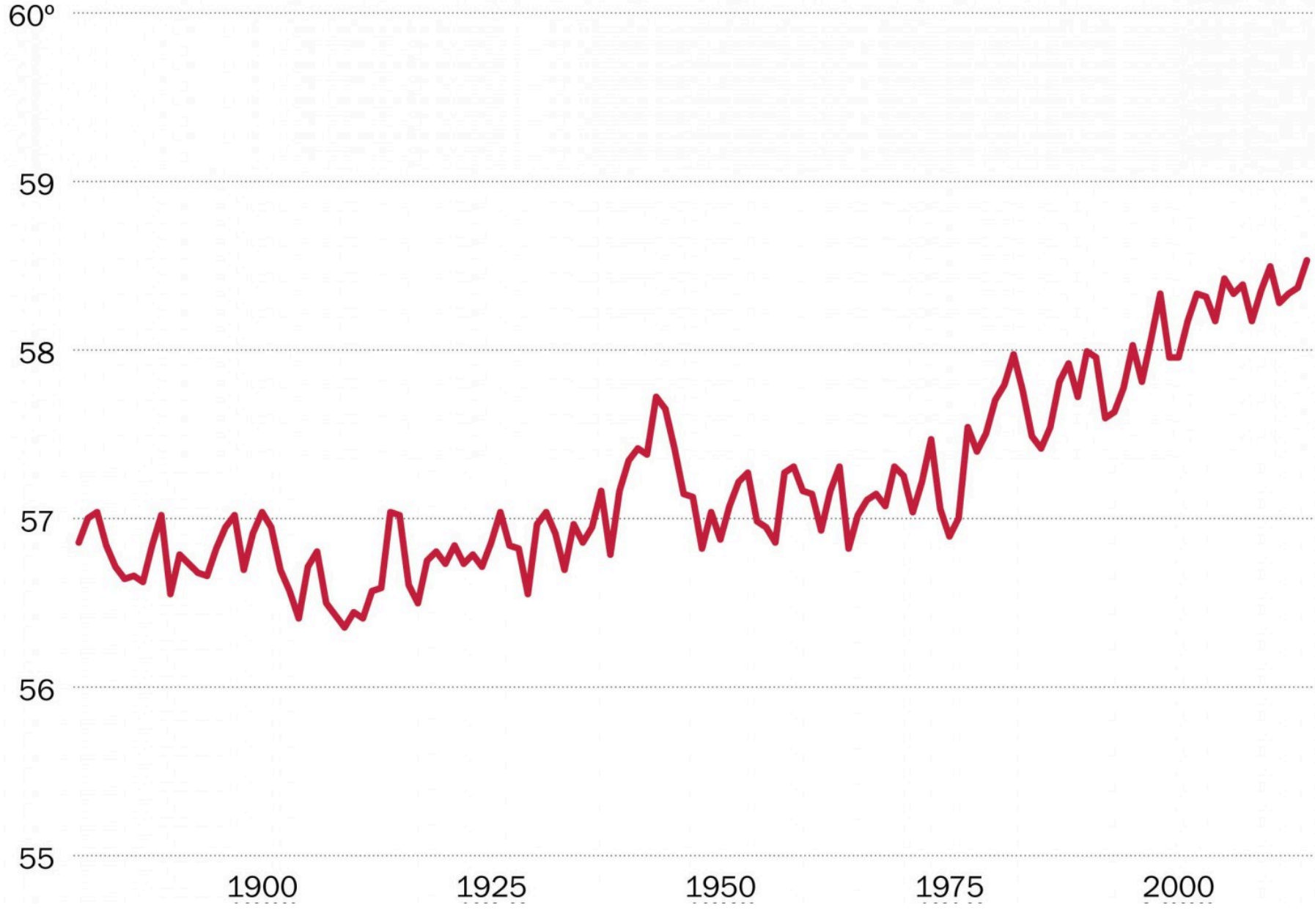
Note: A "household where all parents work" includes single-parent households and dual-earner households. Parents include stepparents and adoptive parents.
Source: Budget Center analysis of US Census Bureau data

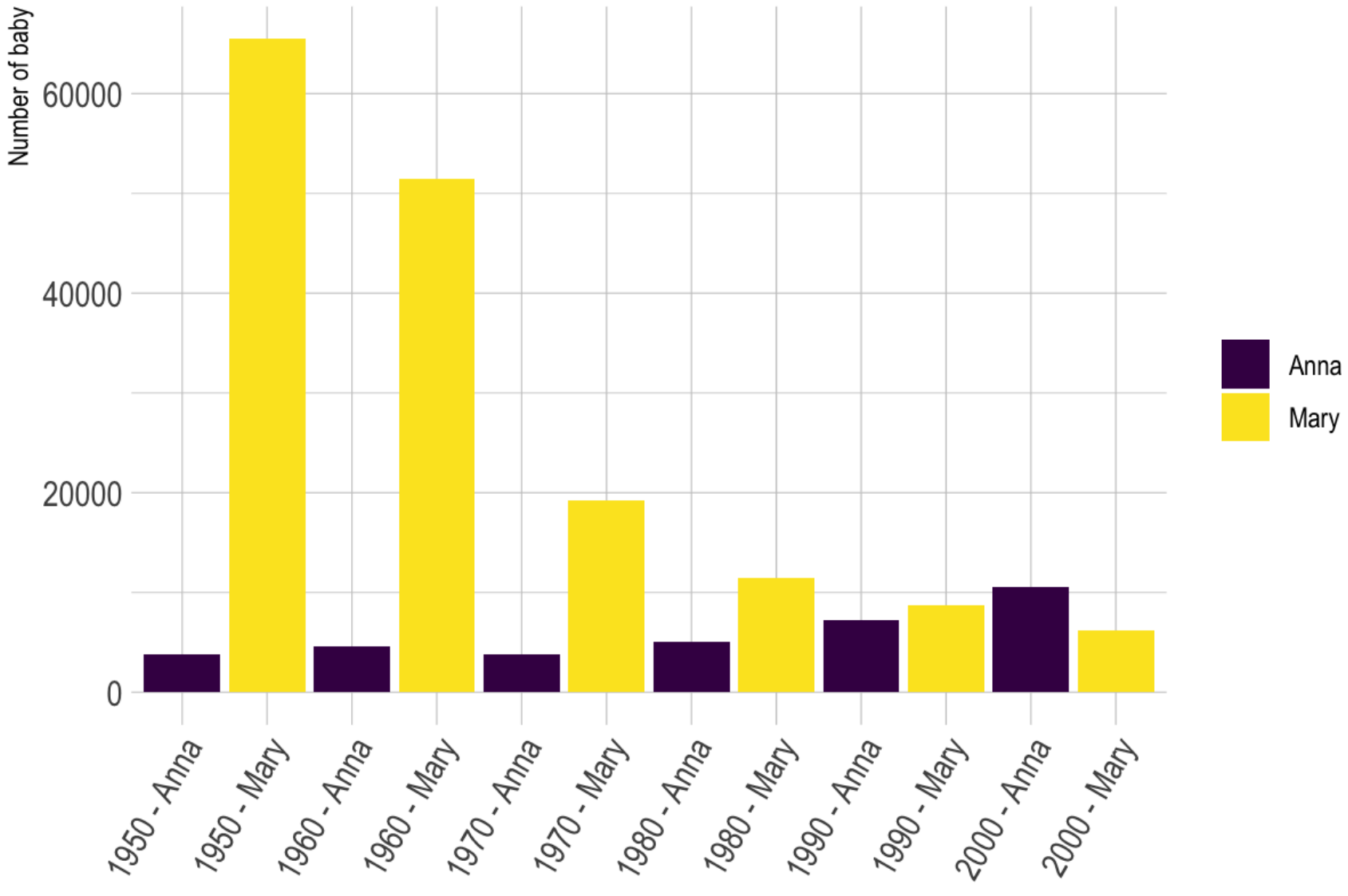
California Budget & Policy Center
Independent Analysis. Shared Prosperity.

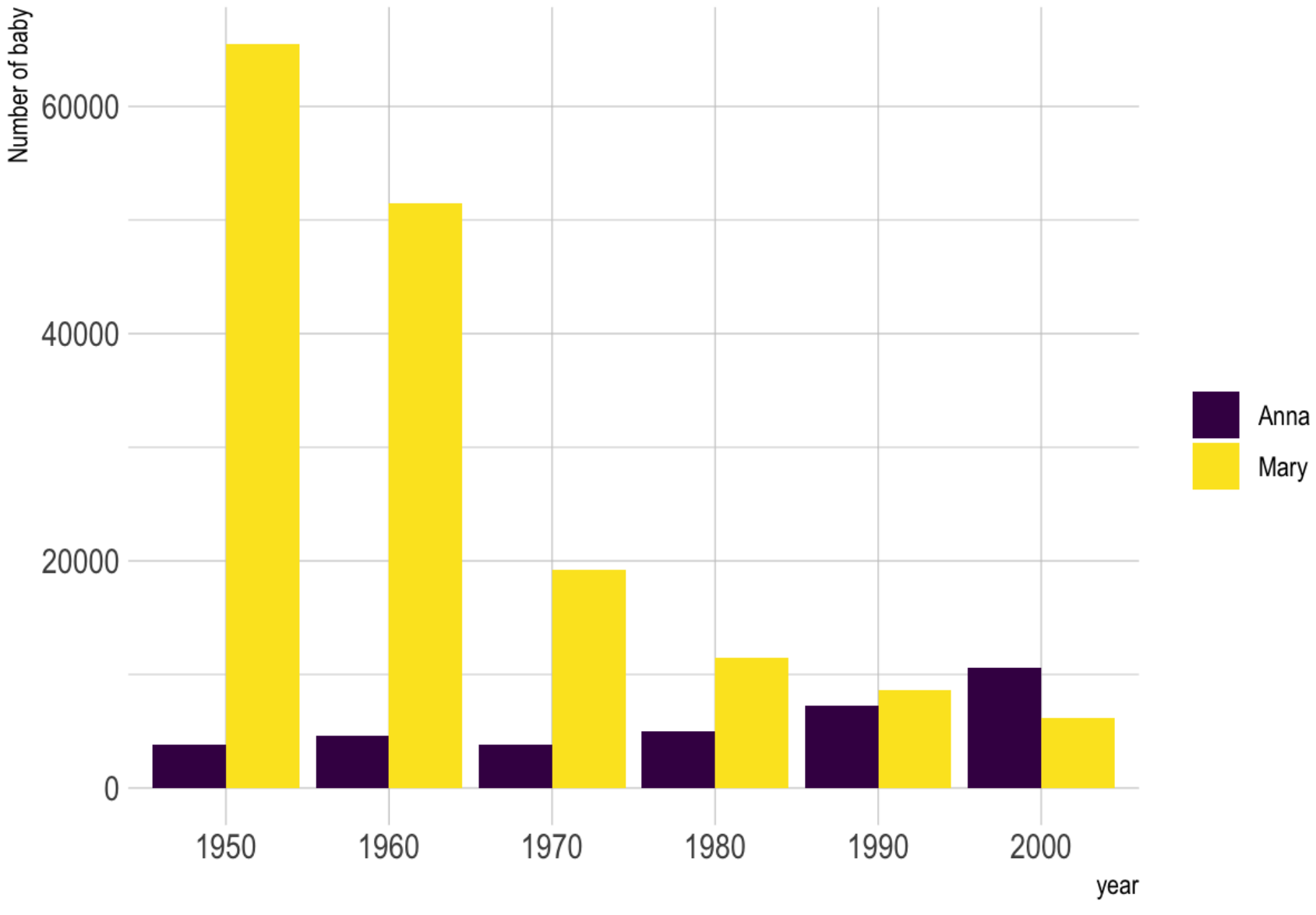

Average Annual Global Temperature in Fahrenheit
1880-2015
110
105
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0
-5
-10
1880
1890
1900
1910
1920
1930
1940
1950
1960
1970
1980
1990
2000
2010

Average global temperature by year

Data from NASA/GISS.




Number of baby
60000
40000
20000
0
1950 - Anna
1950 - Mary
1960 - Anna
1960 - Mary
1970 - Anna
1970 - Mary
1980 - Anna
1980 - Mary
1990 - Anna
1990 - Mary
2000 - Anna
2000 - Mary
Anna
Mary


Number of baby
60000
40000
20000
0
1950
1960
1970
1980
1990
2000
year
Anna
Mary